

**EXPLAINABLE FEATURE DRIFT DETECTION FOR MLOPS
MONITORING**

NUSRAT BEGUM

**A THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF
MASTER OF SCIENCE (COMPUTER SCIENCE)
FACULTY OF GRADUATE STUDIES
MAHIDOL UNIVERSITY
2026**

COPYRIGHT OF MAHIDOL UNIVERSITY

Thesis
entitled
**EXPLAINABLE FEATURE DRIFT DETECTION FOR MLOPS
MONITORING**

.....
Ms. Nusrat Begum
Candidate

.....
Asst. Prof. Thanapon Noraset,
Ph.D. (Computer Science)
Major advisor

.....
Assoc. Prof. Suppawong Tuarob,
Ph.D. (Computer Science)
Co-advisor

.....
Assistant Professor Dr. Warin
Krityakiarana
Deputy Dean for Planning and Quality
Assurance
Acting as Dean of the Faculty of Graduate
Studies, Mahidol University
Dean
Faculty of Graduate Studies
Mahidol University

.....
Assoc. Prof. Boonsit Yimwadsana,
Ph.D. (Electrical Engineering)
Program Director
Master of Science Programme
in Computer Science
Faculty of Information and
Communication Technology
Mahidol University

Thesis
entitled
**EXPLAINABLE FEATURE DRIFT DETECTION FOR MLOPS
MONITORING**

was submitted to the Faculty of Graduate Studies, Mahidol University
for the degree of Master of Science (Computer Science)

on
February 2026

.....
Ms. Nusrat Begum
Candidate

.....
Assistant Professor Dr. Akara Supratak,
Ph.D. (Computer Science)
Chair

.....
Asst. Prof. Thanapon Noraset,
Ph.D. (Computer Science)
Member

.....
Assoc. Prof. Suppawong Tuarob,
Ph.D. (Computer Science)
Member

.....
Assistant Professor Dr. Warin
Krityakiarana
Deputy Dean for Planning and Quality
Assurance
Acting as Dean of the Faculty of Graduate
Studies, Mahidol University
Dean
Faculty of Graduate Studies
Mahidol University

.....
Dr. Pattanasak Mongkolwat,
Ph.D. (Computer Science)
Dean
Faculty of Information and
Communication Technology
Mahidol University

ACKNOWLEDGEMENTS

Nusrat Begum

CONTENTS

LIST OF TABLES

LIST OF FIGURES

Fac. of Grad. Studies, Mahidol Univ.M.Sc. (Computer Science) /

Figure

Page

APPENDIX

APPENDIX A

IMPLEMENTATION DETAILS

To ensure reproducibility, the specific hyperparameters and algorithmic configurations used for the EADD framework are detailed below.

A.1 Adversarial Classifier (LightGBM)

The adversarial classifier utilizes LightGBM (?), a high-performance gradient boosting framework, with the following configuration:

Table A.1: LightGBM Hyperparameters

Parameter	Value
Boosting Type	gbdt
Number of Estimators	50
Learning Rate	0.1
Max Depth	-1
Num Leaves	31
Objective	Binary Classification
Metric	AUC
Verbosity	-1

A.2 Windowing and Monitoring

- **Reference Window (W_{ref}):** 500 samples, maintained via Reservoir Sampling (?).
- **Current Window (W_{cur}):** 200 samples, sliding window.
- **Monitoring Frequency:** Every 50 samples.

A.3 Uncertainty and Statistical Testing

- **Prediction Uncertainty Index (PUI):** Entropy-based warning.
- **Adaptive Sequential Permutation Test (ASPT):** Max 50 perms, $\alpha = 0.01$.

A.4 Drift Severity Index (DSI) Tiers

- **Critical (DSI > 0.6):** concentrated univariate.
- **Moderate (DSI 0.3–0.6):** subset drift.
- **Low (DSI < 0.3):** diffuse shift.

BIOGRAPHY

NAME	Ms. Nusrat Begum
DATE OF BIRTH	21 February 1998
PLACE OF BIRTH	Chattogram, Bangladesh
INSTITUTIONS ATTENDED	Stamford International University, 2017–2022 Bachelor of Business Administration (International Bus Mahidol University, 2024–2026 Master of Science (Computer Science) Mahidol University
E-MAIL	nusrat.beg@student.mahidol.edu

